Introduction to Cross-Validation

Sabyasachi Chatterjee

1 The Basic Problem

Cross-validation is one of the most widely used tools in modern statistics and machine learning. It provides a data-driven way to estimate prediction error and to choose tuning parameters.

Suppose we observe training data

$$D_n = \{(X_1, Y_1), \ldots, (X_n, Y_n)\},$$

and use a learning algorithm A to produce a fitted predictor

$$\widehat{f}_n = A(D_n).$$

For example, A might be ordinary least squares, polynomial regression, ridge regression with a fixed value of λ, a tree-based method, or a neural network with fixed tuning parameters.

For squared error loss, this is

The training error

$$\frac{1}{n}\sum_{i=1}^{n}(Y_i - \widehat{f}_n(X_i))^2$$

is usually too optimistic as an estimate of prediction error, because the same data are used both to fit the model and to evaluate it. As the model becomes more flexible, training error typically decreases, even when prediction error starts to increase.

2 Validation-Set Approach

The simplest idea is to split the data into two parts:

- a training set, used to fit the model;
- a validation set, used to estimate prediction error.

The model is fit on the training set and then evaluated on the validation set. For regression, the validation error is often measured by mean squared error; for classification, it is often measured by misclassification error or another classification loss.

This approach is simple and useful, but it has two important drawbacks. First, the estimate can be highly variable because it depends on the particular random split. Second, the fitted model is trained on only a subset of the data. Therefore the validation error may overestimate the prediction error of the model that would be trained on all n observations.

To address these shortcomings, one would like a method that uses all of the available observations for both training and validation while still providing an estimate of prediction error. Cross-validation was developed precisely for this purpose.

3 K-Fold Cross-Validation

Divide the data into K roughly equal-sized folds

$$C_1, \dots, C_K.$$

Typically, the folds are chosen randomly. For each $k = 1, \dots, K$, fit the learning algorithm on all data except fold C_k, and evaluate the fitted model on the observations in C_k.

For squared error loss, define

$$\mathrm{MSE}_k = \frac{1}{|C_k|} \sum_{i \in C_k} \left(Y_i - \widehat{f}^{(-k)}(X_i) \right)^2,$$

where $\widehat{f}^{(-k)}$ is the predictor trained without fold C_k. The K-fold cross-validation estimate is

$$\mathrm{CV}_K = \sum_{k=1}^{K} \frac{|C_k|}{n} \mathrm{MSE}_k.$$

For classification, one can replace squared error by misclassification loss:

$$\mathrm{Err}_k = \frac{1}{|C_k|} \sum_{i \in C_k} \mathbf{1}\{Y_i \neq \widehat{Y}_i^{(-k)}\},$$

and then average the fold-wise errors.

Common choices are $K = 5$ and $K = 10$. When $K = n$, the procedure is called leave-one-out cross-validation, or LOOCV.

4 Leave-One-Out Cross-Validation

In LOOCV, each fold contains exactly one observation. The LOOCV estimate is

$$\mathrm{CV}_{\mathrm{LOO}} = \frac{1}{n} \sum_{i=1}^{n} L\left(Y_i, \widehat{f}^{(-i)}(X_i) \right),$$

where $\widehat{f}^{(-i)}$ is the model trained with observation i removed.

For ordinary least squares (or any other linear smoother), there is a useful shortcut. If $\widehat{Y}_i$ is the fitted value from the full least squares fit and h_i is the ith diagonal entry of the hat matrix, then

$$\mathrm{CV}_{\mathrm{LOO}} = \frac{1}{n} \sum_{i=1}^{n} \left(\frac{Y_i - \widehat{Y}_i}{1 - h_i} \right)^2.$$

Thus LOOCV can be computed from a single least squares fit.

LOOCV has low bias in the sense that each model is trained on $n - 1$ observations. However, its estimate can have high variance because the training sets overlap heavily. In practice, 5-fold or 10-fold cross-validation often provides a better bias–variance compromise.

5 Bias-Variance Tradeoff in Cross-Validation

Like any statistical estimator, a cross-validation estimate has both bias and variance.

Consider first the validation-set approach. Because the model is trained on only a fraction of the available observations, its prediction error is often larger than the prediction error of the model that would be trained on the entire dataset. Consequently, the validation-set estimate tends to be biased upward.

K-fold cross-validation reduces this bias by training on a larger fraction $(K-1)/K$ of the data. As K increases, the training sets become closer in size to the full dataset, and the bias decreases.

At the extreme, when $K = n$, we obtain leave-one-out cross-validation (LOOCV). Each model is trained on $n-1$ observations, so the bias is typically very small.

However, variance behaves differently. In LOOCV, the training sets for different folds differ by only a single observation. As a result, the fitted models are highly correlated, and the individual error estimates tend to be highly correlated as well. Averaging highly correlated quantities does not reduce variability very much, so the LOOCV estimate can have relatively high variance.

Smaller values of K produce less correlated training sets and often yield a more stable estimate of prediction error, although at the cost of increased bias. These are general rules of thumb, the actual correlations depend on the model in hand and can be complicated to calculate.

Thus there is a bias–variance tradeoff in the choice of K. Large values of K produce low bias but potentially high variance, while small values of K produce higher bias but lower variance. Empirically, $K = 5$ and $K = 10$ often provide a good compromise and are the most common choices in practice.

6 What Quantity is Cross-Validation Estimating?

It is important to distinguish two notions of prediction error.

Once the training data D_n have been observed, the fitted predictor $\widehat{f}_n = A(D_n)$ is fixed. The prediction error of this particular fitted predictor is

$$\mathrm{Err}(D_n) = \mathbb{E}\left[L\{Y, \widehat{f}_n(X)\} \mid D_n\right].$$

This is the *conditional test error*. It depends on the particular training sample that happened to be observed.

Now average over the randomness of the training sample itself:

$$\mathrm{Err} = \mathbb{E}_{D_n}\left[\mathrm{Err}(D_n)\right].$$

Equivalently,

$$\mathrm{Err} = \mathbb{E}_{D_n}\mathbb{E}_{(X,Y)}\left[L\{Y, A(D_n)(X)\} \mid D_n\right].$$

This is the *expected test error*. It is a property of the learning procedure A, rather than of one particular fitted model.

An important conceptual question is: *what quantity is cross-validation actually estimating?* At first sight, one might expect cross-validation to estimate the prediction error of the final fitted model obtained from the observed dataset.

Consider K-fold cross-validation. In each fold, the learning algorithm is trained on a different subset of the data, producing a different fitted predictor. The cross-validation estimate is obtained by averaging prediction errors across all of these different fits. For this reason, it is not obvious that cross-validation is estimating the conditional test error of the final model fitted to the observed dataset. Rather, it behaves more like an estimator of an expected prediction error that averages over many possible training samples.

This interpretation is emphasized in Chapter 7 of ESL Hastie et al. [2009] which notes that cross-validation should be viewed primarily as a tool for estimating prediction risk and comparing learning procedures. More recently, Bates et al. [2023] studied this question carefully in the setting of a linear model and distinguished several notions of prediction error, showing that the precise target of cross-validation depends on the setting and on the form of the procedure being evaluated. Their analysis reinforces the view that cross-validation is best interpreted as evaluating a learning procedure rather than the performance of a single fitted model.

7 Cross-Validation for Model Selection

In practice, cross-validation is most often used for model selection, where one must choose a tuning parameter that controls the flexibility of the fitted model, such as the degree of a polynomial regression, the penalty parameter in ridge regression or the lasso, the complexity parameter of a tree, or the bandwidth of a kernel smoother.

For each value of λ, compute a cross-validation score

$$\mathrm{CV}(\lambda).$$

Then choose

$$\widehat{\lambda} = \arg\min_{\lambda} \mathrm{CV}(\lambda).$$

Finally, after choosing $\widehat{\lambda}$, refit the method on the full dataset using this selected value.

In many applications, the exact numerical value of $\mathrm{CV}(\lambda)$ is less important than the relative comparison across tuning parameters.

8 An Example

Suppose we observe a binary response $Y_1, \ldots, Y_n$ ($n = 25$ say) where each Y_i is Bernoulli(1/2). We also observe $p = 10,000$ predictor variables $X_1, \ldots, X_{10000}$ where every predictor is generated independently of the response and is also Bernoulli(1/2). Thus there is no relationship whatsoever between Y and the predictors. In this situation, no prediction method can perform substantially better than random guessing, and the optimal prediction error is therefore $1/2$.

Now consider the following procedure.

1. Compute the sample correlation between each predictor and the response.
2. Select the 10 predictors having the largest absolute correlations.
3. Fit a logistic regression model using only these 10 selected predictors.

Because there are 10,000 predictors, purely by chance some of them will exhibit large correlations with the response. The resulting logistic regression model may therefore achieve a surprisingly small training error, even though there is no true signal in the data.

Suppose we now perform cross-validation only on the logistic regression step, treating the 10 selected predictors as fixed. Is the resulting estimate of prediction error valid?

The answer is no. The feature-selection step has already used the response values to identify predictors that appear informative. Consequently, the selection step is part of the learning algorithm and must be included in the cross-validation procedure. If the feature selection is performed only once at the beginning, information from the validation folds leaks into the training process, leading to an overly optimistic estimate of prediction error.

The correct approach is to repeat the entire procedure within each fold. For every training fold, one must again select the 20 predictors using only the training data and then fit the logistic regression model using those selected predictors. In other words, cross-validation should be applied to the complete learning algorithm, not merely to the final fitting step.

The general lesson is that any operation that uses the data to construct the final predictor—feature selection, variable screening, tuning parameter selection, dimension reduction, or model fitting—must be repeated inside each cross-validation fold.

9 Theoretical Justification

The theoretical justification for cross-validation is fundamentally frequentist. Cross-validation was introduced by Stone [1974], who showed that it can be viewed as an approximately unbiased estimator of prediction risk. Subsequent work established that, under suitable conditions, cross-validation can consistently estimate the prediction performance of a learning procedure and can be used to select tuning parameters.

For example, suppose that a family of learning procedures is indexed by a tuning parameter λ. Define the prediction risk by

$$R(\lambda) = E\left[L\big(Y, \hat{f}_\lambda(X)\big)\right]. \tag{1}$$

Cross-validation produces an estimate $\mathrm{CV}(\lambda)$ of this risk and selects the tuning parameter

$$\widehat{\lambda} = \arg\min_{\lambda} \mathrm{CV}(\lambda). \tag{2}$$

Let

$$R^\star = \inf_{\lambda} R(\lambda) \tag{3}$$

denote the smallest achievable prediction risk within the family of procedures under consideration. A central question in the theory of cross-validation is whether

$$R(\widehat{\lambda}) - R^\star \longrightarrow 0 \tag{4}$$

as the sample size increases.

In other words, one asks whether the procedure selected by cross-validation achieves asymptotically the same prediction risk as the best procedure that could have been chosen with knowledge of the true risk function. This question has been studied in many settings beginning with the seminal work of Stone [1974], and remains an active area of research today.

The precise answers depend on the learning procedure under consideration. However, the broad conclusion is that cross-validation is justified through repeated-sampling arguments: if the sample size is sufficiently large, the cross-validation estimate behaves similarly to the prediction error that would be observed on new data. For further discussion, see Bates et al. [2023].

References

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